

Parking Detection Method Based on Finite-State Machine and Collaborative Decision-Making

Hongmei Zhu, Shengzhong Feng, and Fengqi Yu

Abstract—A high-accurate parking detection method for wireless magnetic sensor networks is proposed in this paper, which is based on the combination of finite-state machine and collaborative decision-making. The magnetic disturbance induced by a vehicle can be sensed and processed to obtain the availability of a parking space by a magnetic sensor. However, the main challenge lies in the difficulty of eliminating the interferences from adjacent vehicles that decrease the accuracy. The vehicles include moving vehicles on adjacent roads and parking vehicles in adjacent parking spaces. For simplicity and low-energy consumption, a multi-interim finite-state machine (MiFSM) is proposed to deal with the interferences from moving vehicles. Our method contains preliminary detection and final detection. Using MiFSM, most of the parking vehicles can be correctly detected in the preliminary detection. However, the adjacent parking vehicles may cause more complicated interferences. It is hard to distinguish these interferences from the disturbances induced by “weak-magnetic” vehicles above on the detecting sensor. The “weak-magnetic” vehicles cause small magnetic disturbance because of their high chassis or short car-body. Therefore, by using the collaborative information of adjacent sensors, a Dempster–Shafer evidence theory-based collaborative decision-making is developed to cope with these complicated interferences in the final detection. The experimental results show that our work has a significant improvement in detection accuracy, as about 99.8% for vehicle arrival and 99.9% for vehicle departure. The proposed method can also be extended for moving vehicle detection, speed estimation, and vehicle classification in the applications of intelligent traffic system.

Index Terms—Parking detection, vehicle detection, finite state machine, collaborative decision making, WSN.

I. INTRODUCTION

PARKING difficulty becomes one of the serious issues for sustainable urban development with the increased number of cars [1]. The effort to deploy smart parking system (SPS) is always a strategic solution for resolving the parking problem [2]. For administrators, the SPS manages and maximizes the utilization of parking lots. For users, it provides the availability information of parking spaces and dynamic

driving guide. It has been reported that the SPS is efficiently implemented via sensor-based vehicle detectors [3]. These detectors connect with each other through wireless sensor networks (WSNs) and update the status of parking space in real-time [4].

Different types of detectors, such as inductive-loop, ultrasonic, camera, and magnetic sensor, have been used to monitor the status of parking spaces. The inductive-loop detector is a mature technology which is usually installed in traffic intersection and parking entrance [5]. However, it needs to cut road-surface and interrupt traffic which are the obvious drawbacks. The ultrasonic-based solution has many advantages, such as high-accuracy and easy installation. However, it is sensitive to extreme air turbulence and temperature fluctuation [6]. With the development of image processing technology, video detection gains a growing attention in nowadays [7]. Video detection has many special advantages, such as rich information and visibility. However, its performance is affected by light condition that largely limits its application of the SPS. Compared with these sensors, the magnetic sensor merits in low cost, small size, excellent privacy, weather immunity, and easy for large-scale deployment [8]. Numerous research works based on magnetic sensors were proposed for vehicle detection [9]. The methods used in these works can be roughly divided into four categories: threshold based, cross-correlation based, machine learning based, and data fusion based method.

Due to the simplicity and efficiency of the threshold based method, it becomes the most popular one. S. Y. Cheung et al. verified the validity and practicability of the magnetic sensor for vehicle detection [10]. In their work, a total of 330 out of 332 vehicles were detected by using a threshold-based state machine method in the California PATH Program. However, this method is vulnerable, susceptible to noises and baseline drift. To remove baseline drift, an adaptive threshold method was proposed in [11]. Besides, Yang and Lei [12] introduced a fixed threshold state machine method for low-speed vehicle detection with a high accuracy of 99.05%. A magnetic field model based detection method was designed for vehicle detection in [13]. The accuracy was up to 99%. A second adaptive double window vehicle detection algorithm (SA-DWVDA) with an accuracy about 98% was proposed in [14]. Furthermore, Q. Wang et al. proposed a novel adapted threshold state machine algorithm for street parking detection [15]. An accuracy of 97.5% was observed in their work. In [16] a dynamic threshold detection method was provided with an accuracy of 97.92%. Although a number of studies have been done, as described above, the accuracy of

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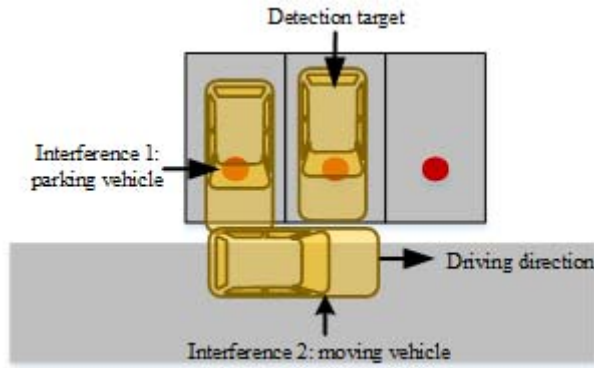


Fig. 1. The two types of interference induced by the adjacent parking vehicle and the moving vehicle respectively.

threshold-based method needs to be improved in order to meet the requirements of real applications.

To improve the accuracy, the cross-correlation based method was presented in several works. Wang *et al.* [17] proposed a cross-correlation method for moving vehicle detection, where a vehicle was modeled as a magnetic dipole and the reference was based on the sensed magnetic disturbance by the vehicle. A reference based on K-means clustering was proposed for parking vehicle detection in our previous work [18]. Recently, a fusion algorithm based on waveform similarity was presented by Zhang *et al.* [19]. Nonetheless, the difficulty lies in obtaining an optimal reference signal which determines the detection accuracy.

In addition, the machine learning based method was also proposed to improve the performance. A signal classification based method, where the status of a parking space was divided into “occupied” and “vacant”, was developed for parking detection [20]. The structure and theoretical model were built in [21]. However, because of the heavy calculation burden, this method is difficult to be implemented on a sensor node.

Moreover, many research works applying sensor fusion technique were proposed to improve the detection accuracy. In order to achieve low-energy consumption, an optical-waked magnetic sensor node for parking detection was provided in [22]. A fusion of magnetic sensor and camera was proposed for vehicle detection in [23]. To achieve high accuracy, fusion method attracts more and more attentions for parking detection [24], [25]. However, this method increases the cost and do not satisfy the need of large-scale deployment in SPS.

We can conclude that the threshold-based method is the most efficient and simplest way implemented on a magnetic sensor node. However, its detection accuracy needs to be improved. The disturbances from adjacent vehicles, including moving vehicles and parking vehicles, may induce mistakes in detection, as shown in Fig. 1. Few publications have taken the interferences into consideration. Therefore, our work focuses on dealing with the interferences from adjacent vehicles.

Our work contains two modules: multi-interim finite-state machine (MiFSM) for preliminary detection and collaborative decision-making based on Dempster-Shafer evidence theory (DS-CDM) for final detection. The state machine is proven to

be both high-efficient and computationally simple [26]. Most of vehicles can be correctly detected because the disturbances are usually strong enough for decision. However, because of the interferences, a few vehicles are missed or false alarmed. So, the D-S theory is used to deal with the uncertainty. The D-S theory is a mathematical theory of evidence proposed by Dempster [27] and developed by G. Shafer in the 1970’s. Due to its effectiveness in processing uncertain information, it has been widely applied to the fields of decision analysis, diagnostics, and statistical inference [28]. Therefore, the combination of MiFSM and DS-CDM is proposed in our work. The main contributions of our paper are listed as following.

First, by eliminating the interferences from adjacent vehicles, our work provides a high-accurate solution without adding any other sensors. It simplifies the system and lowers the power consumption. Secondly, due to the simplicity of the MiFSM and efficiency of the DS-CDM, our method achieves a good balance between the accuracy and energy-consumption. Finally, the performance of this method is immune to the conditions of weather and light. In addition, the convenience of installation and maintenance is also remarkable since there is no need for cutting road-surface and interrupting traffic. The proposed method has been tested in a real parking lot. To ensure the reliability of the proposed method, our test includes different driver behaviors and different types of vehicles.

The remaining of this paper is organized as follows. Section II provides an overview of the proposed parking detection method. The detailed procedure of the combination of MiFSM and DS-CDM are presented in Section III. Section IV describes the hardware setup and testing platform. The experimental verification is presented in Section V. Finally, Section VI contains the conclusion and future research work.

II. THE OVERVIEW OF THE PROPOSED SOLUTION

A. Fundamental Principle of Parking Detection by Using Magnetic Sensors in WSN

All vehicles have significant amount of ferrous metal which can produce a disturbance to the Earth’s magnetic field. This disturbance can be sensed and processed by a magnetic sensor node. In our work, a three-axis magnetic sensor node is used to measure the magnetic field in X, Y, and Z axes simultaneously. As shown in Fig. 2, the X-axis is parallel to the direction of vehicle moving, Y-axis is perpendicular to the X-axis, and the Z-axis is perpendicular to the earth surface and upward.

It was reported that the measurement is related to vehicle’s metallic structure and size [10]. In General, the engine and axle steels make a significant distortion to magnetic field since they contain a great mount of ferromagnetic material, as shown in Fig. 3. So, the waves induced by a same vehicle and measured by different sensors are similar based on a great number of experiments. Therefore, the collaborative information of adjacent sensor nodes can be used for vehicle detection.

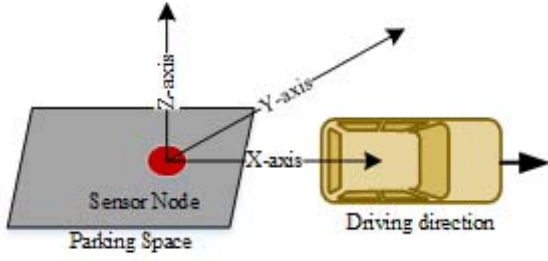


Fig. 2. The direction of vehicle and magnetic sensor node.

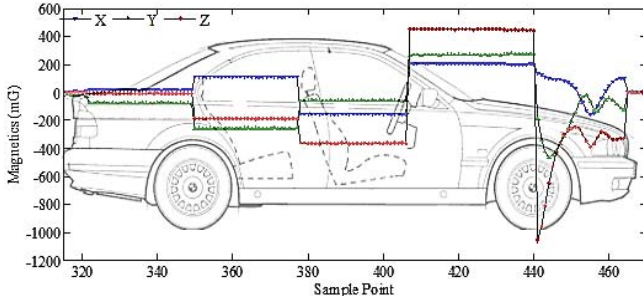


Fig. 3. Different parts of a vehicle cause different distortions to the magnetic field.

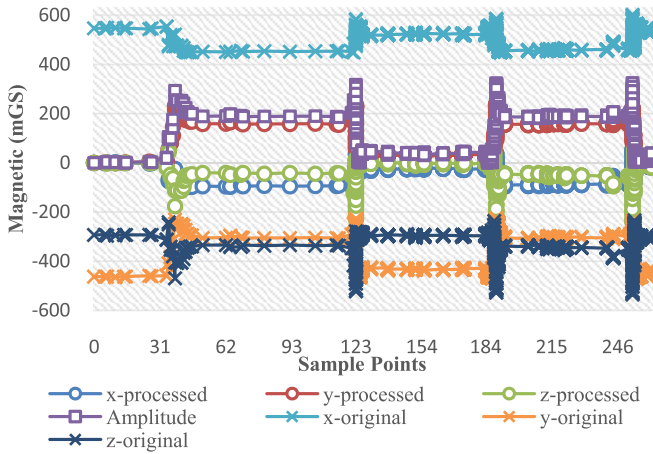


Fig. 4. The original signal and pre-processed signal of magnetic fluctuation sensed by a tri-axial magnetic sensor.

B. Parking Detection Method

In order to eliminate the influence induced by different sensitivity of axis, the signal pre-processing is carried out after data sampling according to Eq. (1). An example of the original signal and pre-processed signal is demonstrated in Fig. 4.

$$M = \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} \quad (1)$$

where, x , y , and z are sampled data in X, Y, and Z axis respectively, x_0 , y_0 , and z_0 are the baseline values accordingly.

Using the pre-processed signal, the parking detection has two steps: MiFSM for preliminary detection and DS-CDM for final detection.

(1) *MiFSM*: Due to the temporary feature, the interferences induced by adjacent moving vehicles can be filtered by a state

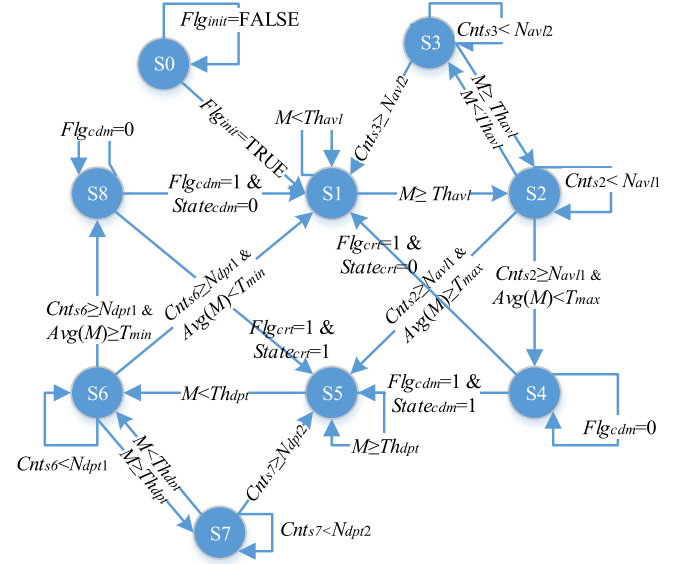


Fig. 5. The state transition diagram of the MiFSM.

machine with several “transient” states. Therefore, four “transient” states have been designed as the interim states in the MiFSM module. In this preliminary detection, most of parking vehicles can be correctly detected. However, the adjacent parking vehicle may cause the complicated interference which is difficult to be distinguished from the disturbance induced by a “weak-magnetic” vehicle above on the detecting node. The “weak-magnetic” vehicles are a few vehicles that cause small fluctuations because of their high chassis or short car-body. Therefore, two “critical” states (*possible-arrival* or *possible-departure*) are designed for these uncertain states. Therefore, the output of MiFSM may be still uncertain or already reach a conclusion. If it is a decided state, the output is considered as the final result. Otherwise, the DS-CDM module is triggered to deal with the uncertainty for final detection.

(2) *DS-CDM*: Taking the advantages of effectiveness and practicability in processing uncertain information, the DS-CDM module based on adjacent sensors is developed to cope with these complicated interferences from adjacent parking vehicles. In this step, the uncertainty results (“critical” states) are transferred to final detections (*occupied* or *vacant*).

III. THE PROPOSED PARKING DETECTION METHOD

A. The MiFSM Module for Preliminary Detection

1) *The State Transition of MiFSM*: A MiFSM method introducing multi interim states is proposed. As shown in Fig. 5, there are nine states in the MiFSM. The definition of each state is listed in Table I.

In brief, at first the MiFSM is started for preliminary detection. All parameters are reset to initial values in **S0**. It is assumed that there is no vehicle parked in the parking space in the initial state. **S1** is the “occupied” state which is a decided state. Because, the interferences from the adjacent moving vehicles usually appear in a short time. The first “interim” state for arrival-decision (**S2**) and for departure-decision (**S6**) are designed for dealing with these interferences.

TABLE I
THE STATES OF MiFSM

State	Action	Description
S0	Initialize, $Flg_{init} \leftarrow 1$	“Initial” state: All parameters are reset to initial values. After initialization, the initialization flag Flg_{init} is set to TRUE.
S1	Status $\leftarrow 0$	“Vacant” state: M is the amplitude of magnetic fluctuation which can be obtained by Eq. (1). Th_{avl} is a pre-defined threshold for arrival-decision.
S2	$Cnt_{S3} \leftarrow 0$, $Cnt_{S2}++$	First “interim” state for arrival-decision: Cnt_{S2} is a counter for counting this S2. N_{avl1} is a pre-defined threshold for counting S2. $\bar{M}$ is the average of fluctuation amplitude. T_{max} is the pre-defined maximum threshold of the “critical” states
S3	$Cnt_{S3}++$	Second “interim” state for arrival-decision: Cnt_{S3} is a counter used for counting S3. N_{av2} is a defined threshold for Cnt_{S3} .
S4	$Cnt_{S2} \leftarrow 0$, $Flg_{cdm} \leftarrow 0$, Status $\leftarrow 2$	“Critical” state for arrival-decision: S4 is defined for dealing with the interferences from adjacent parking vehicles during arrival-decision. Flg_{cdm} is the flag for “critical” state. It goes to S5 or S1 according to the DS-CDM result.
S5	$Cnt_{S2} \leftarrow 0$, $Cnt_{S3} \leftarrow 0$, Status $\leftarrow 1$	“Occupied” state: The parking space is occupied. Th_{dpt} is the threshold for departure-decision.
S6	$Cnt_{S7} \leftarrow 0$, $Cnt_{S6}++$	First “interim” state for departure-decision: Cnt_{S6} is a counter used to count this state. N_{dpt1} is a pre-defined threshold for counting S6. T_{min} is the pre-defined minimum threshold of the “critical” states
S7	$Cnt_{S7}++$	Second “interim” state for departure-decision: Cnt_{S7} is a counter used for counting S7. N_{dpt2} is a defined threshold for Cnt_{S7} .
S8	$Cnt_{S6} \leftarrow 0$, $Flg_{cdm} \leftarrow 0$, Status $\leftarrow 2$	“Critical” state for departure-decision: S8 is defined for dealing with the interferences from adjacent parking vehicles during departure-decision.

TABLE II
STATISTICAL RESULTS FOR PARAMETER SELECTION

Vehicle Type	Number	Arrival-Decision								Departure-Decision							
		max ^a mGS	min ^b mGS	interim ^c mGS	Count ^d	>80 mGS		40~80 mGS		max mGS	Min mGS	Interim mGS	Count	>80 mGS		40~80 mGS	
Minicar	352	788	45	27	5	324	92%	28	8%	697	49	29	5	335	95.2%	17	4.8%
SmallCar	1448	1380	51	29	4	1427	98.5%	21	1.5%	1421	62	28	3	1432	98.9%	16	1.1%
SUV/MPV	579	1075	49	33	6	561	96.9%	18	3.1%	997	55	35	4	559	96.5%	20	3.5%

^a “max” is the maximum of the amplitude induced by the tested vehicles during the arrival or departure.

^b “min” is the minimum of the amplitude induced by the tested vehicles during the arrival or departure.

^c “interim” is the average of the amplitude induced by the tested vehicles in the starting point.

^d “Count” is the average number of “interim” states induced by the tested vehicles.

The second “interim” state for arrival-decision (S3) and for departure-decision (S7) are designed for further reconfirmation due to interferences. During the arrival or departure detection, the short-lived interferences from other vehicles or background noise can be filtered out. Most of vehicles can be correctly recognized as a decided state (S1 or S5).

However, due to the complicated interferences from adjacent parking vehicles, the “critical” states S4 and S8 are defined for arrival-decision and departure-decision, respectively. The “critical” states are uncertain states which is expected to be transferred into decided states through the DS-CDM module.

2) *Discussion of the Parameter Selection:* As described above, the transition conditions of MiFSM depend on several pre-defined thresholds. Therefore, the selection of the thresholds is the key in the design of state-machine.

In order to collect real experimental data, our system is deployed in a commercial parking lot for six years. More than two thousand vehicles have been recorded. The typical magnetic fluctuations caused by 30 vehicles are shown in Fig. 6. The tested vehicles are classified as mini car (Minicar), small car (SmallCar), and sports utility vehicle (SUV). In this work,

2379 vehicles, including 352 minicars, 1448 small cars, and 579 SUVs, have been recorded by our system in the past six years. The statistical results are shown in Table II. Three kinds of information are recorded in Fig. 6.

1) First, the average amplitude of magnetic fluctuations caused by vehicle-arrivals and vehicle-departures is recorded. This information help us select the thresholds T_{max} and T_{min} . In order to distinguish the magnetic disturbance caused by “weak-magnetic” vehicle and interferences, two thresholds T_{max} and T_{min} are defined. From the statistical results in Table II we can see that the amplitudes of the magnetic disturbance caused by most of the vehicles are stronger than 80 mGS. However, a few vehicles cause small fluctuations (40~60 mGS) which are hard to be distinguished from some interferences caused by adjacent vehicles.

As an example shown in Fig. 7, there are 7 disturbances. Disturbances 1 and 2 are induced by the parking vehicles, which are a “weak-magnetic” vehicle and a normal vehicle respectively, on the detected sensor node. The average amplitude of the disturbance 1 is 75 mGS. Disturbances 3~7 are

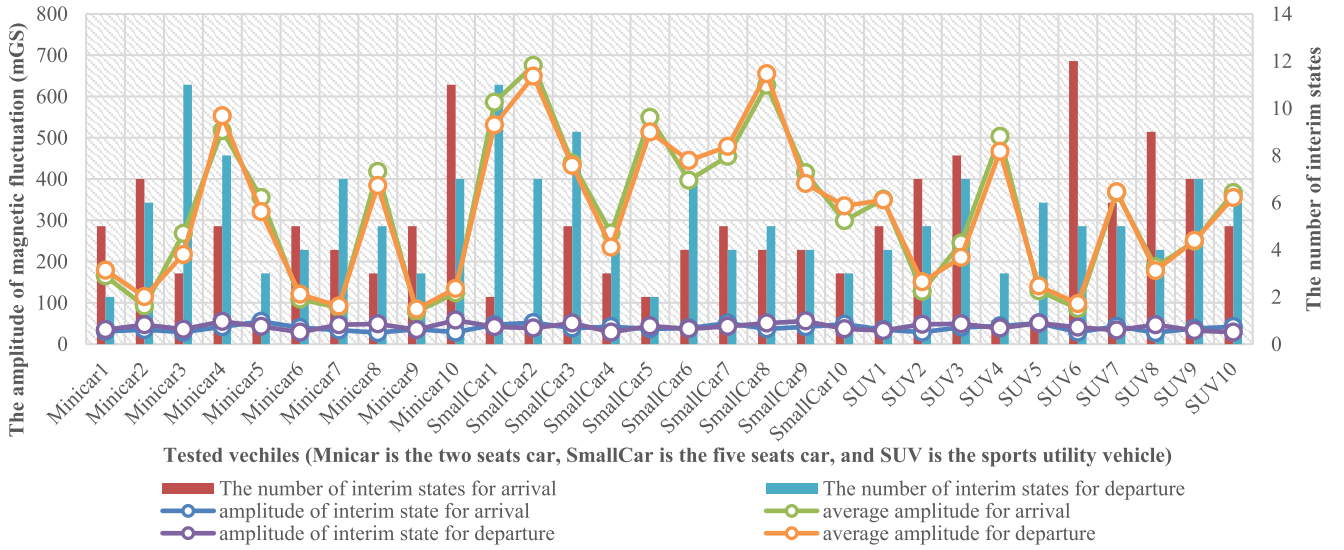


Fig. 6. The typical average amplitude of the magnetic fluctuations caused by 30 vehicles and the number of interim states accordingly.

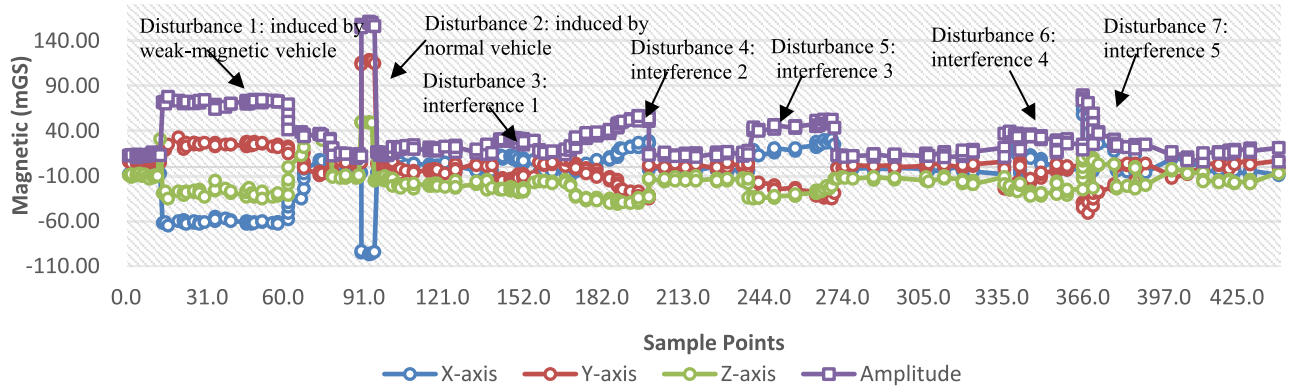


Fig. 7. The average amplitude of magnetic fluctuations caused by 30 vehicles and the number of interim states accordingly.

caused by adjacent vehicles. Among these disturbances, most of them are lower than 40 mGS, e.g. the interferences 1~4. However, if the adjacent vehicle is very close to the sensor node, it can cause a strong disturbance. The amplitude may reach 40~80 mGS, e.g. the interference 7. Therefore we set the $T_{max} = 80$ mGS and $T_{min} = 40$ mGS.

2) Second, the amplitude of magnetic fluctuations caused by interim states, which may be interferences or parking-events, is presented. This information help us choose the thresholds for interim states Th_{avl} and Th_{dpt} . In order to improve the accuracy of parking detection, the interim states are designed for capturing the transient information which is used to suppress the background noises and interferences from moving vehicles. So, Th_{avl} and Th_{dpt} are thresholds for defining the interim states for arrival-decision and departure-decision, respectively. From the statistical results, Th_{avl} and Th_{dpt} are set as 25 mGS.

3) Third, the number of interim states is recorded for selecting the thresholds N_{avl1} , N_{avl2} , N_{dpt1} and N_{dpt2} . These parameters are defined for counting the interim states. And their values are related to the sampling frequency which is set

as 10 Hz. In our work, N_{avl1} , N_{avl2} , N_{dpt1} and N_{dpt2} are set as 5.

B. The DS-CDM Module for the Final Detection

1) *Different Situations for Critical States:* Three situations of “critical” state for possible-arrival on sensor i are taken into consideration, as shown in Fig. 8.

In situation 1 and 3, the “critical” states are caused by the parking vehicle in adjacent parking space $i - 1$ or $i + 1$. The adjacent parking vehicles are a little off the center of their parking spaces and close to sensor i . In these situations, the “critical” states should be transferred to “vacant” state (S1) in the final detection step by using DS-CDM module.

In situation 2, the “critical” state induced by a “weak-magnetic” vehicle above sensor i should be transferred to “occupied” state. Actually, the possibility of two vehicles parking next to each other at the same time is very small. So, the disturbances detected by sensor i and adjacent sensors at the same time are considered to be induced by a same vehicle.

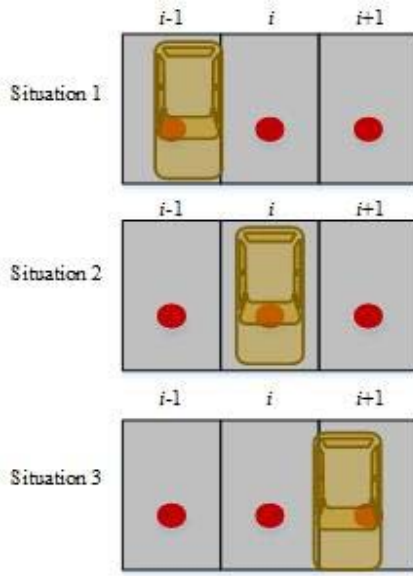


Fig. 8. The parking “critical” states on magnetic sensor i .

The situations of “critical” state for possible-departure on sensor i are similar. The only difference is that the fluctuation is produced by the vehicle departure.

2) *D-S Evidence Theory*: The D-S evidence theory is an efficient method to process uncertain information by fusing several evidences into a new one. As a result, based on the combination of the evidences, a more reliable decision can be made. The theory is based on the nonempty discernment frame which is a finite-set containing all possible answers for a certain problem domain. The “critical” states aim to be transferred to the “occupied” state or the “vacant” state. Therefore, the discernment frame is expressed as $\Theta = \{\text{occupied}, \text{vacant}\}$. Suppose that $E_1, E_2, \dots, E_n$ are n evidences, $m_1, m_2, \dots, m_n$ are n basic probability assignment (BPA) functions resulted from each evidence. The BPA function m_i that is also named mass function with the mapping $2^\Theta \rightarrow [0, 1]$, and $m_i(\phi) = 0$, $\Theta \sum_{x \in \Theta} m_i(x) = 1$. When $\forall x \subseteq \Theta$, D-S combination rule for n mass functions on Θ is shown in

$$P = (m_1 \oplus m_2 \oplus \dots \oplus m_n)(x) = \frac{1}{K} \sum_{x_1 \cap x_2 \cap \dots \cap x_n = x} m_1(x_1) \cdot m_2(x_2) \cdot \dots \cdot m_n(x_n) \quad (2)$$

where, K is the normalization constant and can be obtained by

$$K = \sum_{x_1 \cap x_2 \cap \dots \cap x_n \neq \emptyset} m_1(x_1) \cdot m_2(x_2) \cdot \dots \cdot m_n(x_n) = 1 - \sum_{x_1 \cap x_2 \cap \dots \cap x_n = \emptyset} m_1(x_1) \cdot m_2(x_2) \cdot \dots \cdot m_n(x_n) \quad (3)$$

According to the combination of different evidences on the same discernment frame, the calculated final result can be taken as a basis of decision for uncertainty. In order to deal with the interferences from adjacent vehicles, the collaborative

information from adjacent sensors is provided for final decision making. Because the vehicle arrival and departure have different signal structures, the different evidences of vehicle arrival and departure are introduced as follows.

3) *Evidences for Possible-Arrival*: The discernment frame is expressed as $\Theta = \{\text{occupied}, \text{vacant}\}$. Through observation and analysis of large number of experimental results, four evidences are defined and corresponding mass functions are constructed as following.

1) E_{1a} : The sensor i is the detection target. If there is a vehicle arriving at the parking space i , a significant disturbance will be measured. Otherwise, the disturbance will be relatively weak. Therefore, the amplitude of magnetic fluctuation sensed by sensor i is introduced for the first evidence for arrival detection, E_{1a} . Accordingly, the mass function m_{1a} can be defined as

$$\begin{cases} m_{1a}(\text{occupied}) = \frac{\bar{M}_i - T_{\min}}{T_{\max} - T_{\min}} \gamma \\ m_{1a}(\text{vacant}) = 1 - m_{1a}(\text{occupied}) \end{cases} \quad (4)$$

where $\bar{M}_i$ is the average of pre-processed value described in Eq. (1), sensed by sensor i . The $T_{\max}$ and $T_{\min}$ are the pre-defined maximum and minimum thresholds for “critical” states in the MiFSM, as $T_{\min} \leq \bar{M}_i \leq T_{\max}$. The $m_{1a}(\text{occupied})$ denotes the BPA for “occupied” state by the E_{1a} . In the same way, the $m_{1a}(\text{vacant})$ denotes the BPA for “vacant” state by E_{1a} . Usually, the different mass functions have different contributions to the final decision. Some evidences take more weight on decision. A factor γ is used to adjust the weighting in BPA for E_1 .

2) E_{2a} : The collaborative information between the adjacent magnetic sensor nodes is introduced as an evidence. So, the amplitude of magnetic fluctuation sensed by the adjacent sensor in space $i-1$ is used for the second evidence E_{2a} . Its mass function is described as (5).

$$\begin{cases} m_{2a}(\text{occupied}) = \begin{cases} \frac{\bar{M}_{i-1}}{T_{\min}} \alpha, & 0 \leq \bar{M}_{i-1} < T_{\min} \\ \frac{\bar{M}_{i-1}}{\bar{M}_i} \alpha, & T_{\min} \leq \bar{M}_{i-1} < T_{\max} \\ 0, & \bar{M}_{i-1} \geq T_{\max} \end{cases} \\ m_{2a}(\text{vacant}) = 1 - m_{2a}(\text{occupied}) \end{cases} \quad (5)$$

where $\bar{M}_{i-1}$ is the average of pre-processed value sensed by sensor $i-1$. The ratio of $\bar{M}_{i-1}$ and $\bar{M}_i$ is introduced as the basic probability for “occupied” state. If $0 \leq \bar{M}_{i-1} < T_{\min}$, the sensed signal is lower than the minimum threshold for “critical” states. In this situation, there is no vehicle arrival in space $i-1$. It can be considered as the interference induced by an adjacent parking vehicle or another moving vehicle. If $T_{\min} \leq \bar{M}_{i-1} < T_{\max}$, it is in the range of “critical” state. Therefore, the sensed signal can be caused by an adjacent parking vehicle or a “weak-magnetic” parking vehicle in space $i-1$. If $\bar{M}_{i-1} \geq T_{\max}$, it is stronger than the $\bar{M}_i$. So, it is considered that there may be a vehicle arrival parking in space $i-1$ by this evidence. Thus, the BPA for “occupied” of Space i is set to zero. Similarly, a factor α is used to adjust the weighting in BPA for E_2 and E_3 .

3) E_{3a} : As the E_{2a} , the amplitude of magnetic fluctuation sensed by the other adjacent sensor in space $i + 1$ is used for the third evidence E_{3a} . The mass function can be obtained as

$$\begin{cases} m_{3a}(\text{occupied}) = \begin{cases} \frac{\bar{M}_{i+1}}{T_{min}} \alpha, & 0 \leq \bar{M}_{i+1} < T_{min} \\ \frac{\bar{M}_{i+1}}{\bar{M}} \alpha, & T_{min} \leq \bar{M}_{i+1} < T_{max} \\ 0, & \bar{M}_{i+1} \geq T_{max} \end{cases} \\ m_{3a}(\text{vacant}) = 1 - m_{3a}(\text{occupied}) \end{cases} \quad (6)$$

where $\bar{M}_{i+1}$ is the average of pre-processed value sensed by sensor $i + 1$. As the same as Eq. (5), the ratio of $\bar{M}_{i+1}$ and $\bar{M}_i$ is introduced as the basic probability for “occupied” state.

4) E_{4a} : It has been proven that the signals generated by the same vehicle are similar. So, the cross-correlation of the signal sensed by sensor i and that by an adjacent sensor is introduced for the fourth evidence E_{4a} . The greater fluctuation amplitude between the two adjacent sensors ($i - 1$ and $i + 1$) is chosen for calculating the mass function m_{4a} , as shown in (7).

$$\begin{cases} m_{4a}(\text{occupied}) = \frac{\text{cov}(X_i, X_{adj})}{\sigma_{X_i} \cdot \sigma_{X_{adj}}} \cdot \beta \\ m_{4a}(\text{vacant}) = 1 - m_{4a}(\text{occupied}) \end{cases} \quad (7)$$

where X_i and X_{adj} are the feature vectors sensed by Sensor i and the adjacent sensor respectively. Four features including the number of *peak-value*, *maximum*, *minimum* and *average value* are chosen. The $\text{cov}(X_i \cdot X_{adj})$ is the covariance of X_i and X_{adj} , and $\sigma_{X_i} \cdot \sigma_{X_{adj}}$ is their standard deviation. Like the α, β is a parameter used to adjust the weighting on BPA of E_4 . However, the basic probability for arrival and departure should satisfy the condition: $m_{ia}(\text{occupied}) + m_{ia}(\text{vacant}) = 1$ ($i = 1 \sim 4$).

4) Evidences for Possible-Departure:

1) E_{1d} : As the same as the possible-arrival, the amplitude of magnetic fluctuation sensed by sensor i is introduced for the first evidence for departure detection, E_{1d} . The mass function m_{1d} can be obtained by Eq. (4).

2) E_{2d} : The amplitude of magnetic fluctuation sensed by the adjacent sensor in space $i - 1$ is the second evidence E_{2d} . The mass function m_{2d} can be obtained by Eq. (5).

3) E_{3d} : The amplitude of magnetic fluctuation sensed by the adjacent sensor in space $i + 1$ is the third evidence E_{3d} . The mass function m_{3d} can be obtained by Eq. (6).

4) E_{4d} : The arrival information has been recorded during departure detection. Because the fluctuations recorded in the arrival detection and the departure detection are produced by the same vehicle. Thus, the matching-degree between the arrival and departure signals is used as an evidence for departure decision. The mass function m_{4d} can be calculated as

$$\begin{cases} m_{4d}(\text{occupied}) = 1 - \frac{|\bar{M}_i - \bar{M}'_i|}{\max(\bar{M}_i, \bar{M}'_i)} \\ m_{4d}(\text{vacant}) = 1 - m_{4d}(\text{occupied}) \end{cases} \quad (8)$$

where $\bar{M}_i$ and $\bar{M}'_i$ are the average of fluctuation-amplitudes sensed by sensor i for arrival and departure decision respectively, which are induced by the same vehicle.

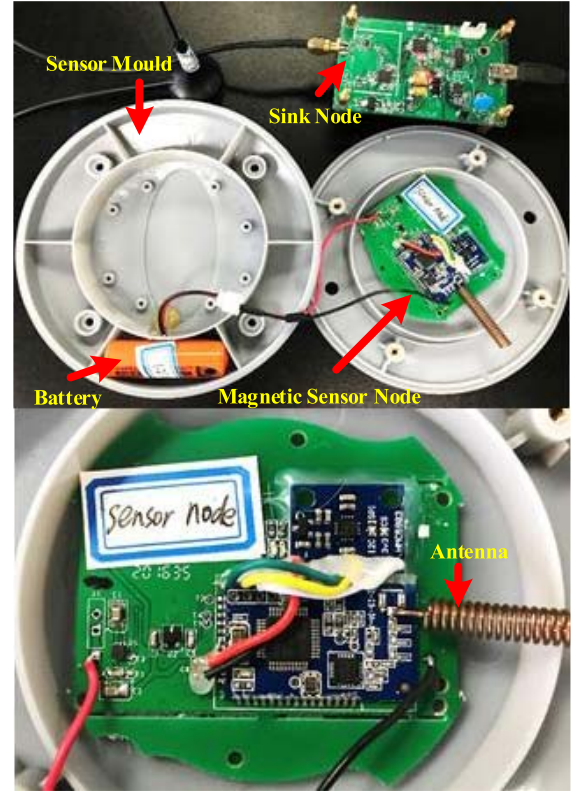


Fig. 9. The magnetic sensor node for parking detection.

5) $DS\text{-}CDM$ for Final Detection: After the BPA of each evidence, a combination result (P) can be obtained by Eq. (2). The final decision is made by (9).

$$\text{Status} = \begin{cases} 1, & P_a(\text{occupied}) \geq \text{TH or } P_d(\text{vacant}) < \text{TH} \\ 0, & P_a(\text{occupied}) < \text{TH or } P_d(\text{vacant}) \geq \text{TH} \end{cases} \quad (9)$$

where the $P_a(\text{occupied})$ is the probability of vehicle arrival for a vacant parking space. Similarly, $P_d(\text{vacant})$ is the probability of a vehicle departure of a occupied parking space. The TH is the defined threshold for the final decision.

IV. HARDWARE AND TEST PLATFORM

A. The Magnetic Sensor Node

The magnetic sensor node is installed in each parking space. All nodes construct a WSN. The Honeywell HMC5983 is chosen to sense the disturbance of the magnetic field. It is a high-resolution magneto-resistive sensor which works well in sensing the Earth's magnetic field. Due to the temperature compensated integrated magnetometer, it provides a high sensitivity under wide operating temperature range. In addition, it is in a small size ($3.0 \times 3.0 \times 0.9\text{mm}$) LCC surface mount package. Moreover, it has low operating voltage (2.16 to 3.6V) and low power consumption. As shown in Fig. 9, besides the magnetic sensor, MCU, RF module, and a battery is also included in our sensor node.

The STM8L151C8 is an ultra-low-power MCU which supports five low-power modes to achieve the best compromise between low-power consumption, short startup. Silicon Laboratories Si4438 is a high-performance and low-power

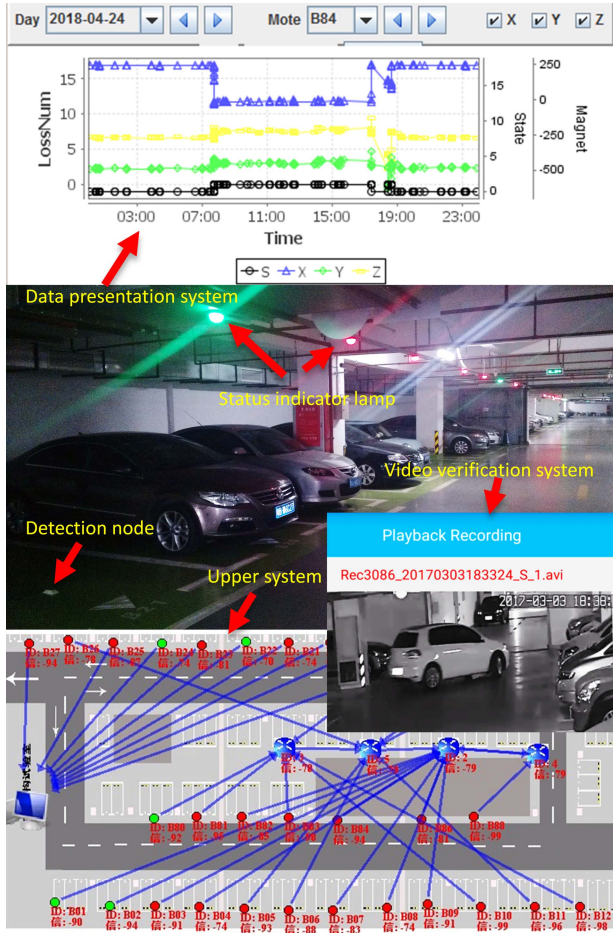


Fig. 10. The test platform used in the parking lot at the SIAT.

dissipation transceiver working in a wide frequency range from 425 to 525 MHz. Our solution provides robust communication links due to the exceptional output power of up to +20dBm with outstanding TX efficiency. In addition, the choice of battery is important since it directly determines the lifetime of the node.

B. The Test Platform

To evaluate the performance, a test platform, including sensor nodes, upper system, indicator, data presentation, and video verification, has been built up in the underground parking lot at Shenzhen Institutes Advanced Technology (SIAT), Chinese Academy of Sciences, as shown in Fig. 10.

Using the test platform, our experiments were carried out. Through WSN, the detection results are sent to base station which is connected with upper computer system. The status of parking space is presented on the upper system that also displays the network information. At the same time, the color of parking indication light is updated to remind the drivers. In order to calculate the accuracy of the proposed method, the data presentation system and video verification system are designed.

V. VERIFICATION

A. Parameter Setup

The parameters are listed in Table III.

TABLE III
PARAMETERS

Symbol	Description	Value
T_{min}	The minimum threshold for “critical” states.	40 mGS
T_{max}	The maximum threshold for “critical” states.	80 mGS
α	Factor for adjusting the weight of BPA of E_2 and E_3 .	0.8
β	Factor for adjusting the weight of BPA of E_4 .	0.8
γ	Factor for adjusting the weight of BPA of E_1 .	1
TH	The threshold for final decision	0.5

TABLE IV
EXAMPLE OF COLLABORATED DECISION

Status	m_{1a}	m_{2a}	m_{3a}	m_{4a}	K	result
occupied	0.98	0.42	0.15	0.86	2.63	0.71
vacant	0.02	0.58	0.88	0.14		0.29

B. An Example for Applying the Proposed Method

An example of applying MiFSM is illuminated in Fig. 11. In the MiFSM module, four arrivals (status = 1) and four departures (status = 0) have been correctly detected by the sensor in Space i . However, a “critical” state (status = 2) occurs. Therefore, DS-CDM module is triggered for final detection. The detection result can be verified by checking the surveillance video.

The $\bar{M}_i = 79$ mGS, $\bar{M}_{i-1} = 21$ mGS, $\bar{M}_{i+1} = 6$ mGS, $X_i = [7, 0.45, 4, 25]$, $X_{adj} = [4, 0.15, 5, 8]$ for the “critical” state. The calculation steps are listed as follows.

(1) The $\bar{M}_i = 79$ mGS, $T_{min} = 40$ mGS, and $T_{max} = 80$ mGS are substituted into the Eq. (4) to calculate the $m_{1a}(o)$ and $m_{1a}(v)$. According to the parameters in Table I, $m_{1a}(o) = 0.98$ and $m_{1a}(v) = 0.02$ are obtained.

(2) Using Eq. (5), the $\bar{M}_{i-1} = 21$ is substituted into $(\bar{M}_{i-1}/T_{min}) * \alpha$, we obtain $m_{2a}(o) = 0.42$ and $m_{2a}(v) = 0.58$.

(3) In the same way, the $\bar{M}_{i+1} = 6$ is substituted into Eq. (5), the $m_{3a}(o) = 0.12$ and $m_{3a}(v) = 0.88$ are obtained.

(4) The $X_i = [7, 0.45, 4, 25]$, $X_{adj} = [4, 0.15, 5, 8]$ are put into Eq. (6), $m_{4a}(o) = 0.86$ and $m_{4a}(v) = 0.14$ are obtained.

(5) Based on the calculated results, the normalization constant $K = 2.63$ can be obtained using Eq. (3).

(6) The final results are obtained by substituting the calculated values into Eq. (2). The results are listed in Table IV. According to Eq. (9), the “critical” state finally reaches the “occupied” state.

C. Verification

The detection accuracy is evaluated by calculating the ratio of the number of parking events correctly-detected and the total number of parking events.

$$Accuracy_j = \left(\frac{CN_j}{TN_j} \right) \times 100\%, j = 0 \text{ or } 1 \quad (10)$$

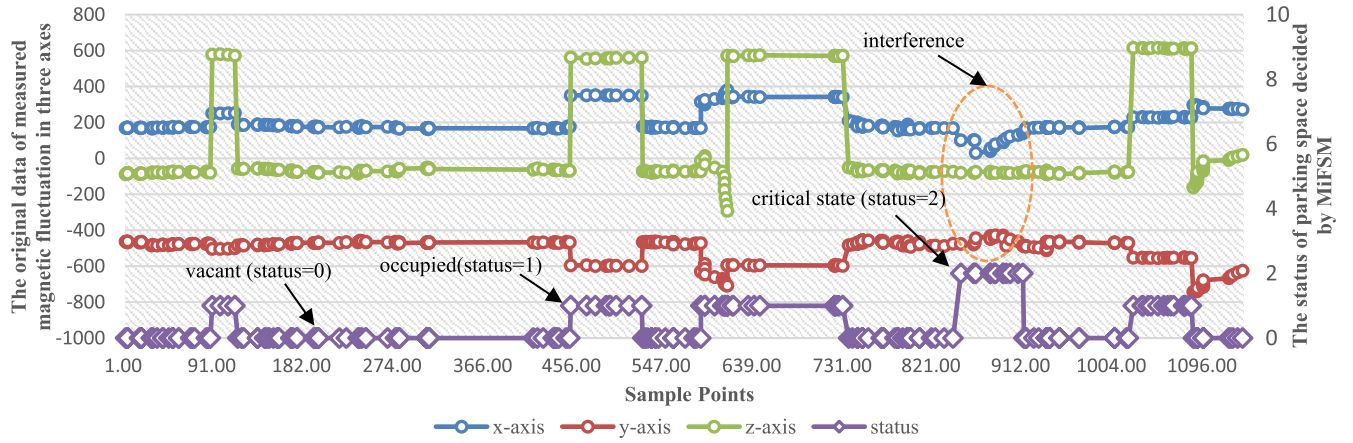


Fig. 11. The example of MiFSM detection. The measurement of magnetic data for three axes and the result of states includes “vacant” state (status = 0), “occupied” state (status = 1), and “critical” state (status = 2).

TABLE V
RESULTS OF COLLABORATIVE DECISION FOR POSSIBLE-ARRIVAL

No	$\bar{M}_i^a$	$\bar{M}_{i-1}$	$\bar{M}_{i+1}$	$m_{1a}(o)^b$	$m_{1a}(v)$	$m_{2a}(o)$	$m_{2a}(v)$	$m_{3a}(o)$	$m_{3a}(v)$	$m_{4a}(o)$	$m_{4a}(v)$	$P_a(occupied)$	$P_a(vacant)$	result
1	45	3	5	0.13	0.87	0.06	0.94	0.1	0.9	0.15	0.85	0.01	0.99	0
2	52	6	167	0.30	0.70	0.12	0.88	0	1	0.89	0.11	0.14	0.86	0
3	75	13	27	0.88	0.12	0.26	0.74	0.54	0.46	0.91	0.09	0.80	0.20	1
4	69	33	4	0.73	0.27	0.66	0.34	0.08	0.92	0.92	0.08	0.71	0.29	1
5	61	29	5	0.53	0.47	0.58	0.42	0.1	0.9	0.87	0.13	0.54	0.46	1
6	78	15	19	0.95	0.05	0.3	0.7	0.38	0.62	0.68	0.32	0.67	0.33	1
7	49	7	214	0.23	0.78	0.14	0.86	0	1	0.83	0.17	0.11	0.89	0
8	62	9	31	0.55	0.45	0.18	0.82	0.62	0.38	0.86	0.14	0.61	0.39	1
9	72	6	8	0.80	0.20	0.12	0.88	0.16	0.84	0.67	0.33	0.36	0.64	0
10	47	120	2	0.18	0.83	0	1	0.04	0.96	0.8	0.2	0.05	0.95	0
11	55	175	5	0.38	0.62	0	1	0.1	0.9	0.85	0.15	0.15	0.85	0
12	51	4	2	0.28	0.72	0.08	0.92	0.04	0.96	0.17	0.83	0.02	0.98	0
13	79	21	6	0.98	0.02	0.42	0.58	0.12	0.88	0.86	0.14	0.71	0.29	1

^aThe unit of $\bar{M}$ is mGS.

^bFor length limitation, the $m_{1a}(occupied)$ and $m_{1a}(vacant)$ are shortened to $m_{1a}(o)$ and $m_{1a}(v)$ in this table.

where the CN_j is the number of correctly detected parking vehicles. The TN_j is the total number of parking vehicles. The $j = 0$ represents arrival detection and $j = 1$ departure detection.

In our test, a total of 1834 vehicles have been tested during two months by using ten magnetic sensor nodes in the under-ground parking lot at the SIAT. The test vehicles consist of minicars, common 4-seat sedan cars, SUV cars, and commercial vans.

As mentioned above, most of the vehicles were correctly detected by MiFSM. There were only 13 “critical” states for possible-arrival and 7 “critical” states for possible-departure. Besides, three out of 1834 vehicles (2 for arrival and 1 for departure) were missed in total. Using DS-CDM in the final detection, all of these “critical” states were transferred to correct decided states and the results have been listed in Table V and VI. In the “result” column, “1” denotes the parking space being “occupied” and “0” denotes “vacant”. The outputs of DS-CDM for possible-arrival and possible-departure are shown in Fig. 12 and Fig. 13, respectively.

In Fig. 12, in the x-axis, the $m_{1a}(o)$, $m_{2a}(o)$, $m_{3a}(o)$, and $m_{4a}(o)$ are the support rate of each evidence for possible-arrival according to Table V. The y-axis represents the resultant of support rate obtained by equation (6) for arrival decision, described as $P_a(occupied)$. In Fig. 13, in the x-axis, the $m_{1d}(v)$, $m_{2d}(v)$, $m_{3d}(v)$, and $m_{4d}(v)$ are the support rate of each evidence for possible-departure according to Table VI. In the same way, the y-axis is the resultant of support rate for departure decision, described as $P_d(vacant)$.

The detection results are shown in Table VII. In the preliminary detection, the accuracy of the MiFSM reaches 98.36% for arrival and 99.13% for departure. The accuracy of arrival and departure are improved to 99.78% and 99.89% for departure using the final DS-CDM detection.

D. Discussion

Comparing with the other parking detection methods based on magnetic sensors, because of the low computational complexity of the state machine, our MiFSM has the advantage of low energy-consumption. Beside, by using the collaborative

TABLE VI
RESULTS OF COLLABORATIVE DECISION FOR POSSIBLE- DEPARTURE

No	$\bar{M}_i^a$	$\bar{M}_i^b$	$\bar{M}_{i-1}$	$\bar{M}_{i+1}$	$m_{1d}(o)^b$	$m_{1d}(v)$	$m_{2d}(o)$	$m_{2d}(v)$	$m_{3d}(o)$	$m_{3d}(v)$	$m_{4d}(o)$	$m_{4d}(v)$	$P_d(occupied)$	$P_d(vacant)$	result
1	78	130	13	40	0.95	0.05	0.26	0.74	0.8	0.2	0.40	0.60	0.72	0.28	1
2	60	120	100	7	0.5	0.5	0	1	0.14	0.86	0.50	0.50	0.12	0.88	0
3	50	170	5	6	0.25	0.75	0.1	0.9	0.12	0.88	0.71	0.29	0.12	0.88	0
4	48	99	3	9	0.2	0.8	0.06	0.94	0.18	0.82	0.52	0.48	0.08	0.92	0
5	69	85	32	26	0.725	0.275	0.64	0.36	0.52	0.48	0.19	0.81	0.54	0.46	1
6	47	121	2	7	0.175	0.825	0.04	0.96	0.14	0.86	0.61	0.39	0.07	0.93	0
7	69	95	33	21	0.725	0.275	0.66	0.34	0.42	0.58	0.27	0.73	0.54	0.46	1

^aThe unit of $\bar{M}$ is mGS.

^bFor length limitation, the $m_{id}(occupied)$ and $m_{id}(vacant)$ are shortened to $m_{id}(o)$ and $m_{id}(v)$ in this table.

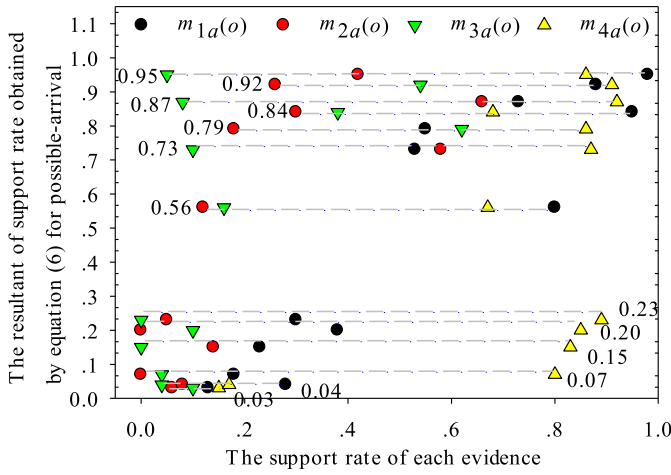


Fig. 12. The final result of DS-CDM for possible-arrival.

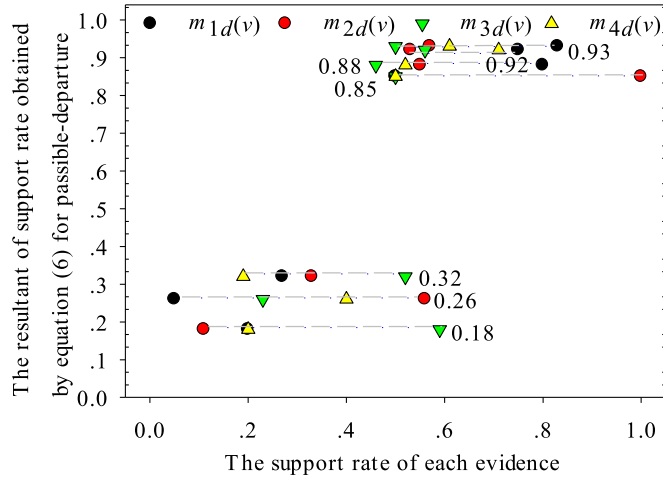


Fig. 13. The final result of DS-CDM for possible-departure.

information, the DS-CDM improves the detection accurate. Although the computational complexity of DS-CDM is a little higher, it runs on upper computer system. Thus, the lifetime of sensor node can reach more than 5 years. Therefore the proposed method achieves dual goals of high accuracy and energy-consumption.

However, because the magnetic sensor is expensive, the cost of this solution is relatively high. With the development of

TABLE VII
DETECTION RESULTS

Event	Total number	Number of preliminary detection	Accuracy of MiFSM	Number of final detection	Final Accuracy
Arrival	917	902	98.36%	915	99.78%
Departure	917	909	99.13%	916	99.89%

sensor technology, the cost may be reduced in the future. In addition, the sensor node is packaged and buried underground in each parking space. The replacement of the node battery is difficult. We can only count on the low-power consumption technology to achieve long lifetime. Moreover, the detection accuracy needs to be further improved.

Very rarely, our system falls into a false negative or false positive state which are caused by two situations: (1) *system malfunction (runaway)*. In this situation, the system should be reset by a remote reset signal which is implemented in our updated sensor node. According our sampled data in the past few years, we have not observed the system malfunction (runaway) happening. (2) *algorithmic problems*. In this situation, our upper system can correct part of the mistakes. This is an issue of detection accuracy. It does not cause to system to halt. The system continues working for next round detection. As for the accuracy of our system, described in Section V, only 3 false negatives out of 1834 tests occurred and they have been corrected by the upper system though reset.

VI. CONCLUSION

In this paper, we have proposed a high accurate parking occupancy detection method by using magnetic sensors. To improve the detection accuracy, our work focuses on dealing with the interferences from adjacent vehicles, including moving and parking vehicles. The performance of the proposed method, which combines MiFSM and DS-CDM together, has been verified in the parking lot at the SIAT (Chinese Academy of Sciences). The experimental results have shown that our method achieves dual goals of high-accuracy and low-power consumption. In the preliminary detection of MiFSM, we have tested 1834 vehicles. The detection accuracy

reaches 98.36% for vehicle arrival and 99.13% for vehicle departure. In the final detection of DS-CDM, 13 “critical” states for arrival and 7 “critical” states for departure are transferred to decided states by using the collaborative information between the adjacent sensor nodes. So, the final detection accuracy reaches 99.8% for vehicle arrival and 99.9% for vehicle departure.

In our future studies, we intend to extend the collaborative decision-making mechanism to moving vehicle detection, speed estimation, and vehicle classification in the application of intelligent traffic system.

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